

Per-Competition Case Studies

Companion to *Benchmarking AI Agents for Kaggle Competitions*

All printed numbers are real measurements as of 2026-07-30.

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Why this document exists

The main report states its claims at the level of methods and evaluation sets; the evidence behind several of them is a handful of competitions examined very closely. That forensic depth belongs neither in the report’s running text, where it pulls the center of gravity toward single competitions, nor in the engineering log, which records what broke while building the machinery. It lives here. Most of these cases concern s3e19, the benchmark’s canonical failure; the fourth records the s5e1 setting behind the pre-registered prediction.

1 The CV–Leaderboard Reversal Mechanism

Detail behind the main report’s analysis *Local CV and the Leaderboard Disagree*. CV and the leaderboard measure different things: CV asks “how well do we do on resamples of the training data,” the leaderboard asks “how well do we do on a batch of never-seen data.” The two correlate strongly when distributions are stable, and decouple as soon as the test–train relationship changes. s3e19 is the extreme case — all three agents’ CVs sit at 10–14 while their leaderboard scores sit at 48–50: everyone misfires together, by amounts unrelated to the CV ordering, into the lower mode of a bimodal leaderboard that CV had no way to predict. The bimodality is sharp in the percentiles: on the 1,174-team leaderboard the 10th percentile is 6.1 while the 25th percentile jumps to 22.8 — almost no middle ground exists between the mode that solved the one-year extrapolation (265 teams below 10) and the mode that did not (570 teams in 40–60).

2 AIDE’s Silent Split Switch, Step by Step

Detail behind the main report’s failure-mode analysis. AIDE’s metric trajectory on s3e19 jumps from 18.42 to 4.41 within a single step:

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step 0–10 : 18.42 ... 22.20 ... 18.42    (time-based validation)
step 11   : 4.41                        <- switched to KFold(shuffle=True)
step 12–19: 4.40, 4.37, 4.40, 4.49      (nine steps polishing a leaked number)
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Step 11’s plan was merely to fix a LightGBM API compatibility issue; while rewriting the training loop it incidentally replaced the validation split with shuffled KFold. The SMAPE definition is identical before and after. This study confirmed the effect with a controlled experiment (same features, same model, split changed only): KFold(shuffle=True) yields SMAPE 4.56, while training on 2017–2020 and validating on 2021 yields 20.41 — a 4.5× gap. The competition’s test set is one full future year; shuffling rows lets the model interpolate within sequences it has memorized. AIDE noticed nothing.

The contrast with my-agent is the sharpest method-level distinction the benchmark surfaced: my-agent also tried shuffled KFold in the same competition (SMAPE 4.2814) but explicitly tagged it in

experiments.json as “DIAGNOSTIC: not used for submission,” its purpose being to isolate whether a score regression came from the split or from the features.

3 The Two Experience-Library Entries Behind the Midterm Correction

Tracing why the midterm had over-attributed s3e19’s lower mode to the missing GDP join, we found the experience library held both readings at once: one entry correctly identified the one-year total extrapolation as the dominant obstacle, while a neighbouring entry over-claimed that joining GDP *dissolves* it. That both coexisted — and that we acted on the optimistic one — is itself a finding: the library’s weak link is retrieval, not content. The over-claiming entry has since been corrected against the leaderboard evidence reported in the main report’s injection-layer experiments.

4 s5e1 (“Forecasting Sticker Sales”): the Setting Behind the Pre-Registered Prediction

playground-series-s5e1 is a fourth firing-class competition in the same country×store×product panel family as s3e19 and tps-sep-2022, with six real countries and a 3-year test horizon against the predecessors’ one year. The proportionality diagnostic read it as the worst violation in the class: cross-country dispersion 0.05–0.21, with the proportionality relation broken in 6 of 7 training years. On that reading — and before any model was fitted or anything submitted — the prediction of the main report’s Result 2b (ratio_target loses to join_feature, by a larger relative margin than on either predecessor) was committed to the competition’s dossier and to version control with its falsifier stated alongside. The three searched arms that then killed it (private MAPE 0.15751 / 0.15626 / 0.12417) applied no remedy for the six violated years: the worst diagnosed violation in the class coincided with the ratio form’s largest win, which is what demoted the diagnostic from a gate to a recorded note.

5 s3e19: the Residual Decomposition in Full

The level/shape split reported in the main report’s Result 3 comes from global rescalings of the best arm’s predictions, labelled as diagnostics and never reported as agent results: ×1.0 scores private SMAPE 48.497, ×1.6 scores 32.932, and ×2.2 scores 46.776 — hence roughly 15.5 points of yearly-level error, with roughly 33 remaining as shape against the leaderboard top of 4.67. The evidence that the needed 1.6 multiplier is unlearnable in sample: the proportionality relation implies 1.05, and the evaluator’s own global-scale grid search settles on 1.0. The transfer test of the experience library’s other candidate recipe: multiplicative panel decomposition, which had beaten a GBDT 4.19 to 8.43 on tps-jan-2022, scores 12.35 on s3e19’s held-out 2021 against the GBDT’s 11.59, and 12.34 even when handed the true held-out yearly total — the recipe fails to transfer even though its proportionality precondition appears satisfied on the training years.